

● RAG Learning

Required Practicals (RP1–RP10)

AQA GCSE Biology (8461) — Paper 1 & Paper 2

Exam Tip: At least 15% of marks in GCSE Biology come from questions about practical skills. Examiners expect you to name equipment, describe safe method steps, identify a named *independent*, *dependent* and *control* variable, and evaluate sources of error. Learn these practicals thoroughly.

RP1 — Microscopy

Aim

Look at both plant cells (e.g. onion epidermis) and animal cells (e.g. cheek cells) under a light microscope, make labelled drawings that include a magnification scale, and estimate the size of the cells.

Method

1. Prepare a slide: peel a thin layer of onion epidermis, add iodine solution and lower a cover slip at an angle (to avoid air bubbles).
2. Start on low power — use the coarse focus first, then switch to high power and use fine focus only.
3. Use an eyepiece graticule (calibrated with a stage micrometer) to measure cell size.
4. Draw a scientific diagram using a sharp pencil — continuous lines, no shading, label with magnification and a scale bar.

Calculations

Magnification = image size $\div$ real size.

Convert between units: 1 mm = 1000 μ m = 1 000 000 nm.

RP2 — Culturing Microorganisms (Biology only)

Aim

Compare how well different antibiotics or antiseptics stop bacteria growing on an agar plate, by measuring the clear zone of inhibition around each one.

Method

1. Sterilise equipment (petri dishes, inoculating loops) by autoclaving or flaming.
2. Work near a Bunsen flame (updraft prevents contamination).
3. Spread a lawn of bacteria on a sterile agar plate using a sterile spreader.
4. Place discs soaked in different antibiotic/antiseptic solutions onto the plate, plus a control disc (water or sterile saline).
5. Secure the lid with a few strips of tape but do not seal it all the way round — oxygen can still get in, which stops harmful anaerobic bacteria growing.
6. Incubate at ≤ 25 °C in schools (to reduce risk of pathogens).
7. After 24–48 h, measure the diameter of the clear "zone of inhibition" around each disc.
8. Calculate area = $\pi \times r^2$ (mm²).

Key Points

- Aseptic technique prevents contamination.
- Larger zone of inhibition = more effective antimicrobial.
- Control disc shows that the effect is caused by the antibiotic, not the paper.

RP3 — Osmosis (Potato)

Aim

Investigate the effect of a range of sucrose/salt concentrations on the mass of plant tissue (potato cylinders).

Method

1. Cut potato cylinders of equal length using a cork borer. Blot and weigh each (initial mass m_1).
2. Place each cylinder into sucrose solutions of known concentration (e.g. 0.0, 0.2, 0.4, 0.6, 0.8, 1.0 mol/dm³).
3. Leave for a fixed time (e.g. 30 minutes) at the same temperature.
4. Remove, blot and reweigh each cylinder (final mass m_2).
5. Calculate the **percentage change in mass** = $((m_2 - m_1) \div m_1) \times 100$.
6. Plot % change in mass (y-axis) against concentration (x-axis). Where the line crosses 0 is the **isotonic point** — the solute concentration of the potato cell sap.

Variables

- **Independent:** sucrose concentration.
- **Dependent:** % change in mass.
- **Control:** potato tissue (same potato), length, time, temperature, volume of solution, blotting.

RP4 — Food Tests

Aim

Identify carbohydrates (sugars and starch), proteins and lipids in food samples with qualitative reagents, including the iodine test (starch), Benedict's test (sugars) and Biuret reagent (protein).

Food type	Reagent	Method	Positive result
Starch	Iodine solution	Few drops on sample	Yellow/brown → blue-black
Reducing sugar	Benedict's	Add to sample, heat in water bath (60–80 °C for 5 min)	Blue → green → yellow → orange → brick-red
Protein	Biuret (alkaline solution + CuSO ₄)	Add to sample at room temperature	Blue → purple/violet
Lipids	Ethanol/Sudan III	Dissolve in ethanol, add to water, or stain with Sudan III	Cloudy white emulsion / red staining

RP5 — Enzyme Activity (Amylase and pH)

Aim

Investigate the effect of pH on the rate at which amylase breaks down starch.

Method

1. Set up a spotting tile with iodine solution in each well.
2. In a test tube, mix a fixed volume of starch, amylase and buffer solution at a chosen pH (e.g. 3, 5, 7, 9, 11).
3. Start a stopwatch; every 30 seconds, take a sample with a pipette and place a drop in the next well (continuous sampling).
4. Record the time taken for the iodine to stop turning blue-black (when all starch is digested).
5. Repeat for each pH and calculate **rate = 1 ÷ time**.
6. Plot rate against pH — a bell-shaped curve peaking at amylase's optimum pH (~7).

Controls

Volumes, concentrations, temperature (water bath at 37 °C), mixing time, and the quality of buffer solutions.

RP6 — Photosynthesis (Light Intensity)

Aim

Investigate the effect of light intensity on the rate of photosynthesis in aquatic plant Cabomba or Elodea.

Method

1. Place a piece of pondweed in a beaker of sodium hydrogencarbonate solution (provides CO₂).
2. Keep temperature constant with a water bath (removes heat from the lamp using a glass beaker as a heat screen).
3. Place a lamp at various distances (e.g. 10, 20, 30, 40, 50 cm) from the beaker.
4. For each distance, count the number of bubbles produced per minute (or collect O₂ in an inverted syringe).
5. Plot rate against 1/d² (which is proportional to light intensity).

Variables

- **Independent:** distance from lamp (→ light intensity).
- **Dependent:** bubbles/min or volume of O₂.
- **Control:** same plant, CO₂ concentration, temperature, time.

RP7 — Reaction Times

Aim

Plan and carry out your own investigation into how one factor (e.g. caffeine or distraction) changes human reaction time, using the ruler drop test.

Method

1. Sit the subject with their arm resting so only their hand hangs off the table.
2. Hold a 30 cm ruler vertically with the 0 cm mark just above their thumb and index finger.
3. Without warning, drop the ruler. The subject catches it as fast as possible.
4. Record the distance (d) the ruler fell.
5. Convert to time using $d = \frac{1}{2}gt^2$, so $t = \sqrt{(2d/g)}$ where $g = 9.8 \text{ m/s}^2$.
6. Vary one factor at a time (e.g. with/without caffeine, with/without distraction) keeping others constant.
7. Repeat many times and calculate means.

Controls

Same ruler, same person/starting position, time of day, temperature.

RP8 — Plant Responses (Biology only)

Aim

Investigate the effect of a factor (light or gravity) on the growth of newly germinated seedlings.

Method

1. Germinate cress seedlings on moist cotton wool in three separate dishes.
2. Keep one dish in all-round light (control), one in unilateral light (from one side), one in darkness.

3. Measure the length of each seedling with a ruler at the start and again each day for several days.
4. Observe the direction of shoot growth and measure the angle of any bending.
5. Record results as length measurements **and** as accurate, labelled biological drawings of the seedlings.
6. For gravity: pin germinating seeds (e.g. broad beans) to damp cotton wool in a dish kept in the dark, with some seeds turned on their side, and observe the direction in which the roots and shoots grow.

Expected Results

- Control (all-round light) — shoots grow straight up.
- Unilateral light — shoots bend towards the light (positive phototropism).
- Darkness — seedlings grow tall and thin (etiolation), pale in colour.
- Gravity — roots grow downwards (positive gravitropism) and shoots grow upwards (negative gravitropism).

Explanation

Auxin is produced at the shoot tip. In unilateral light, auxin moves to the shaded side and causes more elongation there, bending the shoot towards the light.

RP9 — Field Investigations (Sampling)

Aim

Measure the population size of a common species in a habitat, and study the effect of a factor on the distribution of this species.

Random Sampling with Quadrats

1. Measure the total area of the field/habitat.
2. Use random numbers to generate coordinates; place a 1 m × 1 m quadrat at each.

3. Count the number of individuals (or % cover) in each quadrat.
4. Repeat at least 10 times. Calculate mean per m².
5. Multiply by total area to estimate the total population.

Transects (for a gradient)

1. Lay a tape measure along a line crossing the habitat (e.g. from shade into full sun).
2. Place quadrats at regular intervals along the line.
3. Count/estimate cover at each interval.
4. Plot a kite diagram or line graph to show how abundance changes with distance.

RP10 — Decay (Biology only)

Aim

Find out how temperature affects the rate at which fresh milk decays, by following the fall in its pH.

Method

1. Add a fixed volume of fresh milk to a test tube plus a few drops of phenolphthalein indicator (pink in alkali).
2. Add a set number of drops of sodium carbonate solution until just pink.
3. Add lipase enzyme and start a stopwatch.
4. Record the time taken for the solution to turn colourless. Lipase breaks down fats in the milk into fatty acids and glycerol, so the pH falls; phenolphthalein turns colourless once the pH drops below about 8.3.
5. Repeat at different temperatures (e.g. 10, 20, 30, 40, 50 °C) using water baths.
6. Calculate rate = $1 \div \text{time}$.

Interpretation

Rate should increase with temperature up to enzyme optimum; above that, lipase denatures and rate falls.

General Practical Skills

Evaluating an Investigation

- **Accuracy** — how close to the true value.
- **Precision** — how close repeats are to one another.
- **Reliability** — repeats give similar results; take a mean to reduce impact of random error.
- **Validity** — only one independent variable is being changed.
- **Anomalies** — results that stand out; check for procedure errors, ignore when calculating means.

Safe Working

- Wear eye protection with any chemicals; tie back long hair near Bunsens.
- Report breakages of glassware; clean spills immediately.
- Never eat or drink when handling microbes or chemicals.
- Wash hands after practicals involving microbes or biological material.

Using a Microscope

- Low power first, then high power.
- Use the fine-focus control only on high power.
- Draw a scale bar and state the magnification.